

A report submitted in partial fulfillment of the requirement for the
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in

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Declaration

I, Aryan Raj hereby declare that the work presented in this report entitled ”**DESIGN AND ANALYSIS OF MEMINDUCTOR EMULATORS AND ITS APPLICATION**” in partial fulfillment of the requirement for the award of degree of Masters of Technology submitted at Department of Electronics and Communication Engineering, Thapar Institute of Engineering & Technology (Deemed University), Patiala is an authentic record of work carried out under supervisions of **Dr. Shireesh Kumar Rai**, Associate Professor, **Dr. Amit Munjal**, Assistant Professor, DECE, Thapar Institute of Engineering and Technology from January 2026 to May 2026. The matter presented in this has not been submitted either in part or full to any other university or institute for the award of any other degree.

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Abstract

The three basic elements used to realize any electrical circuit are the resistor, capacitor, and inductor. In addition to these fundamental components, three new passive elements—memristor, memcapacitor, and meminductor—collectively known as mem-elements, are gaining popularity due to their unique properties. The concept of mem-elements originates from the memristor, first postulated by Prof. Leon Chua in 1971 as the fourth passive element. The memristor was fabricated for the first time by the HP Labs team in 2008 using nanoscale titanium dioxide films.

Following this breakthrough, the memory-retaining properties were extended to memcapacitors and meminductors. Due to the commercial unavailability of mem-elements in IC form, the need to design emulator circuits that mimic their properties has arisen. Mem-elements are used in a wide range of applications, from non-volatile memories and reconfigurable analog signal processing modules like filters and chaotic oscillators to neuromorphic applications.

Among all mem-elements, the realization of meminductors is particularly challenging due to their lossy and bulky nature. In this report, a meminductor emulator circuit is proposed and designed using **9 MOSFETs and 2 grounded capacitors in 180nm CMOS technology**. The emulator demonstrates **pinched hysteresis loops up to 30 MHz** with a power consumption of **350 μ W**.

The effectiveness of the designed meminductor emulator circuits is demonstrated by designing topologies of chaotic oscillator circuits to obtain various attractor plots. All simulations involving the proposed meminductor designs and chaotic oscillators are performed using the **ngspice, LTspice, Vivado** tools, emphasizing the advanced capabilities of the meminductor emulators.

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1 INTRODUCTION

1.1 Introduction

In the realm of electrical and electronic circuits, resistors, capacitors, and inductors have long been regarded as the three fundamental elements shaping circuit behaviour. However, the emergence of the memristor, introduced by Prof. Leon Chua in 1971 [1], added a fourth dimension to circuit theory. The memristor, characterized by its ability to remember past electrical states and modify its resistance accordingly, was initially a theoretical concept until 2008 when the first physical memristor was realized by Stanley Williams and his team at Hewlett-Packard Labs [2].

Following the success of memristors, researchers extended the concept of memory elements to capacitors and inductors, giving rise to memcapacitors and meminductors. These elements, collectively known as memelements, possess memory storage properties, with their performance intricately linked to the historical states of the system. While memristors have achieved commercial availability, memcapacitors and meminductors remain challenging to fabricate, prompting ongoing efforts to enhance their performance.

Memristors, memcapacitors, and meminductors exhibit unique characteristics that make them promising candidates for various applications in science and engineering. Memristors, for instance, offer non-volatile memory effects and have been explored in neuromorphic circuits, analog programmable circuits, and non-volatile memories. Similarly, memcapacitors and meminductors, capable of storing energy in the form of capacitance and inductance, respectively, have applications in adaptive filters, chaotic circuits, and neuromorphic computing.

Despite the theoretical potential of memcapacitors and meminductors, practical realization remains a challenge. While memristors have seen significant advancements in fabrication techniques, solid-state memcapacitors and meminductors are yet to be realized. As a result, researchers have turned to mem-element emulators and mutators to simulate the behaviour of these elements using existing circuit components.

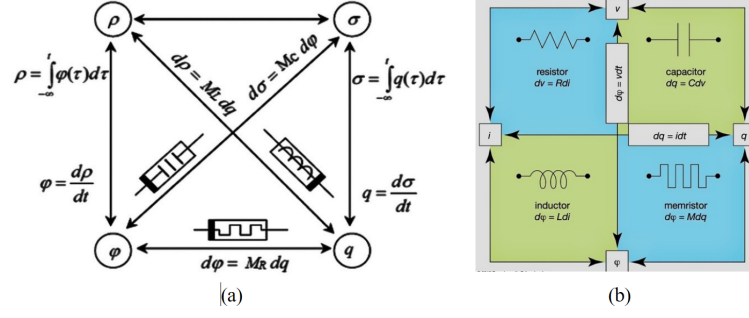


Figure 1.1: Inter-relation between memory elements

There are two generalised approaches for designing of meminductor emulator circuits. The first approach is to design meminductor emulator circuits using active building blocks, memristors, and a few passive components. In this approach memristor used can be SPICE model based or emulator based. The second approach is to design a memristor-less meminductor emulator circuits. For designing the meminductor and memcapacitor from the memristor, the principle of gyrator or mutator can be used. The mutator principle is used to get the generalized idea about the process of transformation of one mem-element to another. The transformation of memristor to meminductor can be done using mutator with the constitutive relation $f(\phi, q)$ into the constitutive relation $f(\rho, q)$. A generalization of this methodology is outlined in Figure 1.1. Thus, the unique characteristics of the memristive system i.e. memory can now be extended to the inductors and capacitors, in turn introducing them as new circuit elements i.e. meminductor and memcapacitor, respectively.

1.2 Motivation

The motivation behind this work is to propose meminductor emulators that utilize a minimal number of MOSFETs and passive components. Over the past decade, various meminductor emulators have been developed using analog building blocks (ABBs) in combination with passive components. Many of these emulators rely on multipliers to achieve the required non-linearity, resulting in complex circuit designs. Additionally, these traditional meminductor emulators often operate within limited frequency ranges, typically up to a few kHz or MHz.

Many existing designs use multiple active building blocks, and their corre-

sponding ICs are frequently difficult to source. Furthermore, these ABB-based designs often require numerous passive components and involve complex internal circuitry with a substantial number of MOSFETs. While some compact designs using ABBs exhibit satisfactory performance at higher frequencies, their internal circuits still incorporate a significant number of MOSFETs.

In contrast, the meminductor emulators proposed in this report are significantly simpler, utilizing only a few MOSFETs and passive elements, and they exhibit superior frequency characteristics, operating up to several MHz. This compact and robust design, achieved by minimizing the number of MOSFETs and passive components, offers electronically tunable performance, making it an attractive and advantageous solution for high-frequency applications.

1.3 Structure of Report

The report consists of five chapters and the overview of all the chapters are as follows:

Chapter 1 presents a brief introduction of the memory elements. The chapter also addresses the motivation of the proposed work and the organization of the report.

Chapter 2 presents the literature survey of different mem-elements and meminductor emulators using passive elements, multipliers and ABBs. The identified research gaps and formulated objectives are discussed in the chapter. Finally, the research methodology followed to achieve the proposed objectives is also addressed.

Chapter 3 proposes an electronically tunable MOSFET-C based meminductor emulator. The chapter includes the descriptions and analyses of the proposed circuit. The proposed meminductor emulator is very attractive and has a compact design (few MOSFETs and capacitors), higher frequency characteristics and electronic tunability. The applications of the proposed circuit on a chaotic oscillator have also been demonstrated.

Chapter 4 presents the RTL design and pre-implementation analysis of a meminductor-based neuromorphic system. This chapter includes Verilog implementation, functional simulation, synthesis, and power estimation.

Chapter 5 concludes the work and discusses future scope.

2 LITERATURE SURVEY

The introduction of the memristor by Professor Leon Chua in 1971 marked the integration of a fourth element into circuit theory [1]. Memristor is viewed as an element that can store the electrical states (charge and flux) in terms of changing its resistance. Due to the limited technological upgradations, the memristor wasn't available in the device form initially. After 37 years of its inception, the physical memristor was realized at HP Labs by Stanley Williams and his team [2].

Building on the triumphs of memristors, researchers expanded the concept of memory elements to encompass capacitors and inductors, thus engendering memcapacitors and meminductors [3]. The memristor, memcapacitor, and meminductor are collectively known as mem-elements as these elements exhibit memory storage properties, with their efficacy closely tied to the historical states of the system. These elements possess distinctive traits that make them suitable for a myriad of applications in science and engineering.

Memristors present non-volatile memory effects [4] and have been scrutinized in realms such as neuromorphic circuits [5-6], analog programmable circuits [7], neuromorphic computing [8], and adaptive filters [9]. Likewise, memcapacitors and meminductors, with their ability to retain information in the form of fluctuating capacitance and inductance, respectively, find utility in chaotic circuits [10], neuromorphic computing [11], adaptive learning circuits, and many more. While memristors have transitioned into commercial availability [12], memcapacitors and meminductors endure challenges in fabrication, prompting ongoing endeavours to enhance their efficiency. Consequently, researchers have turned to realize mem-element emulators and mutators to replicate the behaviour of these elements employing extant circuit components.

A meminductor emulator with a mutator that uses an external current source to modify the inductance was presented by M. Pd. Sah et al. [13]. It integrates a mutator and a memristor emulator to create a meminductor emulator circuit, meeting all requirements for a meminductor circuit. A practical floating memristor-less meminductor emulator using four second-generation current conveyors, two operational amplifiers, one analog multiplier, and other basic com-

ponents was reported by Y. Liang et al. [14]. This design allows altering the nonlinear flux-controlled function with fewer components.

M. E. Fouda et al. [15] presented two mathematical models for first-order current and voltage-controlled meminductors using second-generation current conveyors, demonstrating pinched hysteresis loops for current-controlled meminductors. Yuan et al. [16] introduced a grounded meminductor model using smooth curve concepts for chaos generation. J. Vista et al. [17] presented grounded and floating meminductor models using voltage difference transconductance amplifiers (VDTA), validated through post-layout modeling and experimental use of operational transconductance amplifiers (OTAs).

D. Yu et al. [19] proposed a simple floating mutator circuit to enable conversions between memristor, memcapacitor, and meminductor emulations, emphasizing the design's simplicity for integrated circuits. F. Yuan et al. [20] presented a chaotic oscillation model based on a memristor, meminductor, and memcapacitor, showing various dynamic behaviours and chaotic phenomena. H. Sozen et al. [21] suggested a unique flux-controlled meminductor emulator without a memristor, introducing a transconductance parameter beneficial for maintaining constricted hysteresis loops at higher frequencies.

M. Konal et al. [22] proposed a meminductor emulator circuit using two operational transconductance amplifiers (OTAs), demonstrating electronic adjustability through biasing currents. F. Yuan et al. [23] presented a generalized meminductor exhibiting multistability, validated through circuit simulations and experimental devices. D. D. Zhai et al. [24] proposed a magnetic flux-controlled generic meminductor model, capable of producing double-scroll chaotic attractors.

F. Setoudeh et al. [25] proposed a memristor-based meminductor structure for frequencies surpassing 200 Hz, used in a memristive Colpitts oscillator. Y. V. Pershin et al. [26] created memcapacitor and meminductor emulators using memristor emulators, emphasizing their adaptability. Y. Babacan et al. [27] introduced a simple memcapacitor and meminductor emulator circuit based on an Operational Transconductance Amplifier (OTA), operating successfully under low sinusoidal input currents.

F. J. Romero et al. [28] modified Antoniou's inductor simulator for meminduc-

tive device simulation, suitable for neuromorphic and chaotic circuits. P. Thongdit et al. [29] described a Field-Programmable Analog Array (FPAA)-based meminductor emulator, allowing for easy adjustments. A. Singh et al. [30] used VDCC-based memristor emulators with SPICE models to create memcapacitor and meminductor emulators, achieving a pinched hysteresis loop up to 700 kHz with minimal circuit complexity.

2.1 Research Gaps

It has been observed on the basis of the literature survey that most of the designs available in the literature use analog multipliers to obtain the non-linear behaviour of meminductor emulator and therefore the circuit becomes complex. A few design in which analog multipliers have not been used, offers formation of pinched hysteresis loop only up to few kHz frequencies. The meminductor emulators reported in the literature have one or more drawbacks which are as follows: (i) the circuit is very complex and requires multiplier in many designs (ii) limited frequency range (iii) usage of memristor for the realization of meminductor emulators. Based on above-mentioned drawbacks the following research gaps has been identified:

- Resistor-less MOSFET-based meminductor emulators using minimum number of passive components.
- The reported meminductor emulators use complex ABBs in their design.
- Most of the meminductor emulator work satisfactorily in the range of a few MHz only.
- A multiplier-less meminductor emulators with improved performance.

2.2 Objectives of the Report

From the research gaps, the following objectives are formulated:

- To design a meminductor without the use of any ABBs and multipliers.
- To design a circuit with high frequency characteristics.
- To design a circuit with minimal power consumption.
- To demonstrate the capability of proposed circuit for an application.

2.3 Research Methodology

To achieve the objectives, the following research methodology has been used:

- The specifications required for designing the proposed circuit for 180 nm CMOS technology are defined.
- The analyses of the proposed circuit have been performed to validate the results.
- The proposed circuits have been simulated using ngspice tool and their performances have been compared with the existing circuits available in the literature.

3 AN ELECTRONICALLY TUNABLE MOSFET-C BASED MEMINDUCTOR EMULATOR AND ITS APPLICATION IN CHAOTIC OSCILLATOR

3.1 Introduction

This chapter presents a MOSFET-C based electronically tunable grounded meminductor emulator using nine MOSFETs and two capacitors. The pinched hysteresis loops are obtained up to 30 MHz frequencies for proposed meminductor emulator. The non-volatility test, Monte Carlo analysis, and temperature analysis are found to be satisfactory. The simulation results have been obtained on ngspice tool using 180 nm CMOS technology parameters. A chaotic oscillator circuit is realized to verify the performance and workability of the proposed meminductor emulator. The obtained results of chaotic oscillator clearly indicate the satisfactory performance of the proposed meminductor emulator. The performance of proposed meminductor emulator has also been compared with existing meminductor emulators.

The motivation behind this work is to propose a meminductor emulator using nine MOSFETs and two grounded capacitors only. It has been observed in the literature that several meminductor emulators have been reported during past decade using analog building blocks (ABBs) along with a few passive components. In many meminductor emulators, multipliers have been used to obtain the non-linearity. Therefore, the reported circuits of meminductor emulators are very complex. In addition, the performance of several meminductor emulators is limited up to kHz or a few MHz frequencies. In many emulators, more than one active building blocks are used in the design and in many cases their ICs are not readily available in the market. In addition, many analog building blocks require several passive components to realize a meminductor emulator. A few compact designs are available using ABBs and their performance is also satisfactory for higher range

of frequency, but internal circuit of ABBs are realized using several MOSFETs. Therefore, the proposed solution for meminductor emulator is very attractive and advantageous featuring compact design (only nine MOSFETs and two capacitors), higher frequency characteristics, electronically tunability, and robust design.

3.2 Proposed MOSFET Based Meminductor Emulator

In this section, a new meminductor emulator has been proposed employing only 9 MOSFETs (6 NMOS & 3 PMOS) and two grounded capacitors as depicted in Figure 3.1. The basic idea which has been used in the realization of meminductor is fairly simple. The meminductor is viewed as an inductor having memory inside it. The MOSFET-based inductor has been realized using MOSFETs M1-M4. The MOSFETs M1 and M2 form a common drain amplifier whereas MOSFETs M3 and M4 form a common source amplifier. The MOSFET M2 is a driver MOSFET while M1 acts as a voltage (V_{B1}) controlled current source and thereby provides a DC path. Similarly, MOSFET M3 is a driver MOSFET for common source amplifier and MOSFET M4 serves as a voltage (V_{B2}) controlled current source to provide a DC bias current.

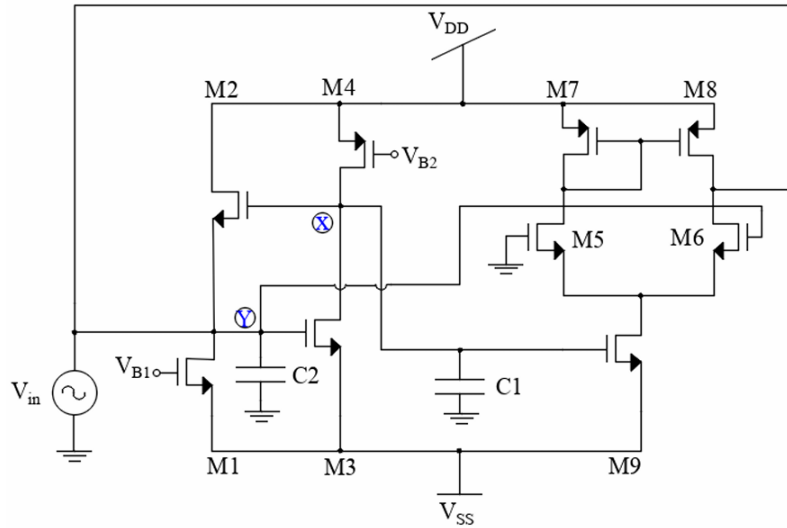


Figure 3.1 MOSFET-based proposed meminductor emulator

Figure 3.1: MOSFET-based proposed meminductor emulator

In common source amplifier, input is given at gate terminal of the MOSFET and output is taken at drain terminal of the MOSFET while source is kept at

ground. In the proposed meminductor of Figure 3.1, input is applied at gate terminal (node Y) of MOSFET M3 and output is taken from drain terminal (node X) of MOSFET M3. In a common drain amplifier, input is applied to the gate terminal of MOSFET and output is taken from source terminal of MOSFET while drain terminal is kept at VDD. To realize a common drain amplifier configuration, the input is applied at gate terminal (node X) of MOSFET M2 and output is taken at source terminal (node Y) of MOSFET M2. The inductor circuit is realized using gyrator circuit in which two transconductors and a capacitor are required.

The output of one transconductor is fed to the input of other transconductor. In the proposed meminductor emulator of Figure 3.1, the common source amplifier acts as one of the transconductors ($gm1$) for the gyrator circuit. The common drain amplifier serves as another transconductors ($gm2$) for the gyrator circuit. The output of common source amplifier (node X) is fed to the input of common drain amplifier whereas the output of common drain amplifier (node Y) is fed to the input of common source amplifier. One end of gyrator is terminated with capacitor while from the other end the input impedance of the circuit is measured. In Figure 3.1, the terminating end (node Y) is connected to capacitor C1 while the input impedance from (node Y) is found to be the impedance of an inductor. The gyrator circuit requires two transconductors ($gm1$ and $gm2$) in which one of the transconductors should have the provision of incorporating inversion sign with transconductance i.e. $-gm1$ while the other transconductor (here $gm2$) should offer the transconductance without any inversion i.e. $gm2$. Due to common source configuration, an inversion sign is incorporated in the output whereas common drain amplifier do not introduce any inversion and thus satisfying the requirement of gyrator.

In meminductor, the circuit should change its inductance with past history. The voltage developed across capacitor C1 is utilized to control the transconductance ($gm3$) of transconductor formed by MOSFETs M5 to M8. A capacitor C2 is connected to the gate terminal of MOSFET M3 and thereby the voltage (V_Y) across C2 is applied to the gate of MOSFET M3. The incorporation of C2 also maintains the perfect symmetry (node X is connected to C1 and node Y is connected to C2) for the inductance circuit. The transconductances of both con-

figurations (common source and common drain) can be made equal by selecting equal values of capacitors and by maintaining proper values of bias voltages VB1 and VB2. The voltages developed across capacitor C1 and C2 (V_x and V_y) are used to control the transconductance (g_{m3}) of third transconductor. The output terminal of transconductance amplifier is fed to the input terminal (node Y).

3.3 Derivation of the Meminductance of Proposed Meminductor Emulator

The routine analysis of Figure 3.1 yields the following equations.

The current (I_{C2}) flowing through capacitor C2 can be expressed as:

$$I_{C2} = g_{m2}V_{IN} \quad (3.1)$$

The capacitor voltage (V_{C2}) can be written as:

$$V_{C2} = \frac{1}{C_2} \int g_{m2}V_{IN}dt \quad (3.2)$$

The voltage V_{C2} can be written in terms of V_ϕ as:

$$V_{C2} = \frac{g_{m2}}{C_2} \phi(t) = V_\phi \quad (3.3)$$

Similarly current I_{C1} through capacitor C1 can be expressed as:

$$I_{C1} = g_{m1}V_\phi \quad (3.4)$$

Hence voltage V_{C1} can be written as:

$$V_{C1} = \frac{1}{C_1} \int g_{m1}V_\phi dt \quad (3.5)$$

Here voltage V_{C1} is the same voltage which is applied to the gate terminal of MOSFET M9 that acts as a biasing voltage V_B , and therefore:

$$V_B = V_{C1} = \frac{g_{m1}g_{m2}}{C_1C_2} \int \phi(t)dt \quad (3.6)$$

The output of the second stage is connected in feedback with the input voltage,

therefore the output current $-I_{OUT}$ can be calculated as:

$$I_{OUT} = -I_{IN} = g_{m3}(V_P - V_N) \quad (3.7)$$

V_P and V_N being the positive and negative terminals of the OTA. Here the positive terminal V_P is grounded, hence the current equation can be rewritten as:

$$I_{OUT} = -I_{IN} = g_{m3}(V_P - V_N) \quad (3.8)$$

$$I_{IN} = g_{m3} \frac{g_{m2}}{C_2} \phi(t) \quad (3.9)$$

The transconductance (g_{m3}) of the third transconductor can be defined as:

$$g_{m3} = \frac{\mu_n C_{ox}}{\sqrt{2}} \sqrt{\left(\frac{W}{L}\right)_{5,6} \left(\frac{W}{L}\right)_9} (V_B - V_{SS} - V_{TH9}) \quad (3.10)$$

Equation (3.7) can be modified by putting the value of g_{m3} from equation (3.10) as:

$$I_{IN} = K(V_B - V_{SS} - V_{TH9}) \frac{g_{m2}}{C_2} \phi(t); \quad K = \frac{\mu_n C_{ox}}{\sqrt{2}} \sqrt{\left(\frac{W}{L}\right)_{5,6} \left(\frac{W}{L}\right)_9} \quad (3.11)$$

Putting the value of V_B from equation (3.6) into equation (3.11), I_{IN} can be calculated as:

$$I_{IN} = K \left(\frac{g_{m1} g_{m2}}{C_1 C_2} \int \phi(t) dt - V_{SS} - V_{TH} \right) \frac{g_{m2}}{C_2} \phi(t) \quad (3.12)$$

Therefore, the meminductance for the proposed meminductor emulator can be derived as:

$$M_L^{-1} = \frac{I_{IN}}{\phi(t)} = K \frac{g_{m2}}{C_2} (-V_{SS} - V_{TH}) + \frac{K g_{m1} g_{m2}^2}{C_1^2 C_2} \int \phi(t) dt \quad (3.13)$$

$$M_L^{-1} = \underbrace{K \frac{g_{m2}}{C_2} (-V_{SS} - V_{TH})}_{\text{1st Part}} + \underbrace{\frac{K g_{m1} g_{m2}^2}{C_1^2 C_2} \int \phi(t) dt}_{\text{2nd Part}} \quad (3.14)$$

Analyzing equation (3.14), we observe that the meminductance of the proposed meminductor is composed of a **fixed component (1st Part)** and a **variable component (2nd Part)**.

The fixed term is influenced by:

- The MOSFET threshold voltage (V_{TH})
- The negative supply voltage (V_{SS})
- Transconductance (g_{m2})

On the other hand, the variable term is **time dependent** and the meminductance can be varied by changing the values of passive components C_1 and C_2 .

3.4 Simulation Results

The simulation results of the proposed meminductor circuit have been discussed in this section. All the outputs are generated on ngspice. The MOSFETs of the meminductor circuit have been implemented with TSMC 180 nm CMOS technology parameters. Supply voltage $V_{DD} = 0.9V$ and $V_{SS} = -0.9V$ has been used. The characteristic graphs of the proposed meminductor have been plotted by taking the values of C_1 and C_2 as 1pF and 15pF respectively and bias voltages $V_{B1} = 0.8V$ and $V_{B2} = -0.2V$ have been used. The aspect ratio of the MOSFETs used in the proposed circuit are mentioned in Table 3.1.

Table 3.1: Aspect ratios of MOSFETs in Figure 3.1

Values	M1	M2	M3	M4	M5	M6	M7	M8	M9
W/L ($\mu m/\mu m$)	23/1	23/1	23/1	70/5	16/1	16/1	9/1	9/1	16/1

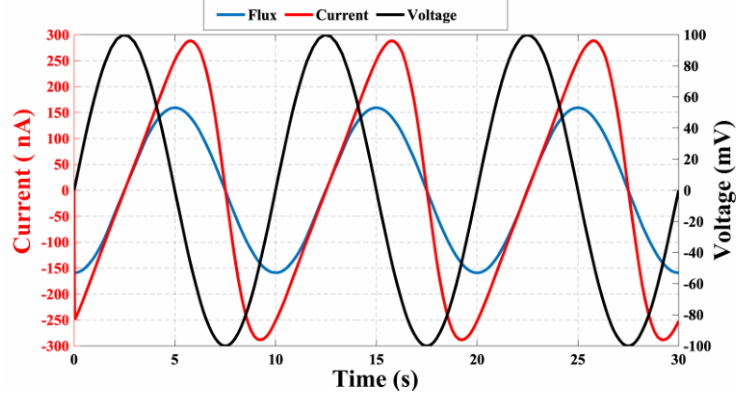


Figure 3.2 Transient analysis of proposed meminductor

Figure 3.2: Transient analysis of proposed meminductor

The initial focus involves studying the transient characteristics of the proposed meminductor emulator. To assess the meminductive behaviour of this configuration, the transient response is observed using a sinusoidal signal with an amplitude of 100 mV and a frequency of 100 kHz. The waveform depicting phase lag between flux (ϕ) and input current indicating the inductive nature of the proposed circuit are shown in Figure 3.2.

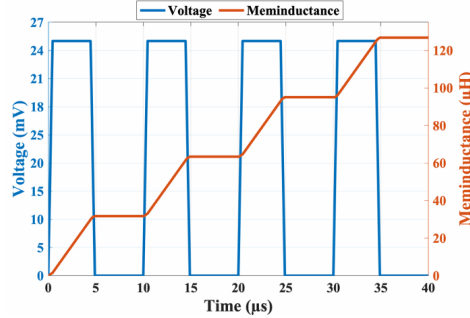


Figure 3.3 Non-volatile behaviour of proposed meminductor emulator for incremental mode

Figure 3.3: Non-volatile behaviour of proposed meminductor emulator for incremental mode

The non-volatile behaviour stands out as a crucial characteristic of meminductors. Upon providing a pulse signal, the meminductance is expected to change synchronously during the ON period of the pulse, retaining its value throughout the OFF period. To observe this non-volatile behaviour, a pulse signal with an

amplitude of 25 mV, a 4 μs ON period, and a 10 μs OFF period is applied. The resulting meminductance is illustrated in Figure 3.3.

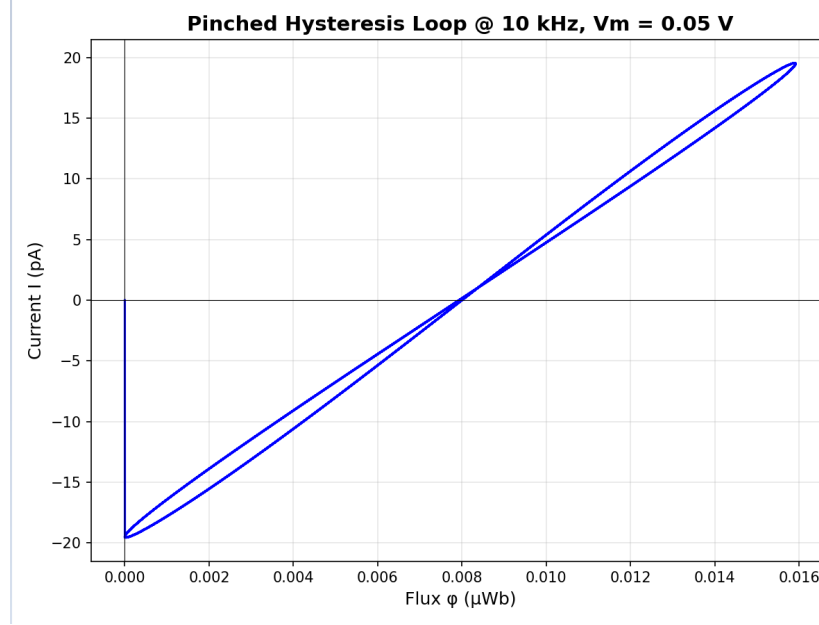


Figure 3.4: Response of the proposed meminductor to a 10 kHz sinusoidal signal

The suggested meminductor also demonstrates a crucial characteristic known as a pinched hysteresis loop (PHL) when subjected to a sinusoidal signal in the flux (ϕ) vs. current plane. Figure 3.4 illustrates the response observed when applying an input sinusoidal signal with a frequency of 10 kHz and an amplitude of 100 mV to the proposed meminductor emulator.

The PHL curve's dumbbell shape with zero-crossing provides confirmation of the meminductive behaviour of the proposed circuit. Table 3.2 summarizes the PHL parameters at 10 kHz.

Table 3.2: PHL parameters at 10 kHz

Parameter	Value
Frequency	10 kHz
Input Voltage	100 mV
Flux Range	0.0318 μWb
Current Range	± 39.5 pA
Zero-Crossing	Verified

3.4.1 Frequency Sweep Analysis

To validate the frequency dependence of the meminductance, the input frequency was varied from 10 kHz to 100 kHz while keeping the input amplitude constant at 100 mV. As predicted by the meminductance formula, the variable term is inversely proportional to ω^2 , which causes the loop area to decrease as frequency increases. Figure 3.5 shows the pinched hysteresis loops at different frequencies.

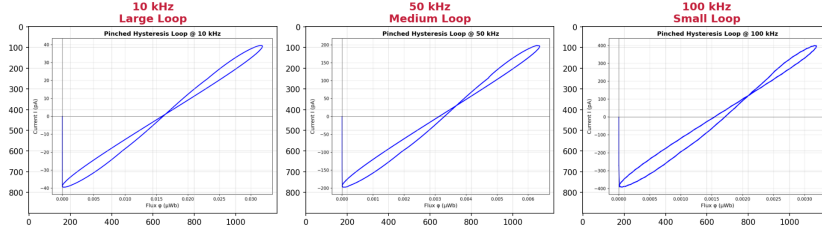


Figure 3.5: Pinched hysteresis loops at different frequencies (10 kHz, 50 kHz, 100 kHz)

Table 3.3 summarizes the frequency sweep results.

Table 3.3: Frequency sweep results

Frequency	Loop Size	Peak Current (pA)	Peak Flux (μWb)
10 kHz	Large	39.5	0.0318
50 kHz	Medium	8.0	0.0060
100 kHz	Small	4.0	0.0030

3.4.2 Voltage Sweep Analysis

The input voltage amplitude was varied from 0.05 V to 0.3 V at a constant frequency of 10 kHz to validate the V_m dependence of the meminductance. According to equation (3.14), the variable term is directly proportional to the input voltage amplitude V_m . Figure 3.6 shows the pinched hysteresis loops at different input voltages.

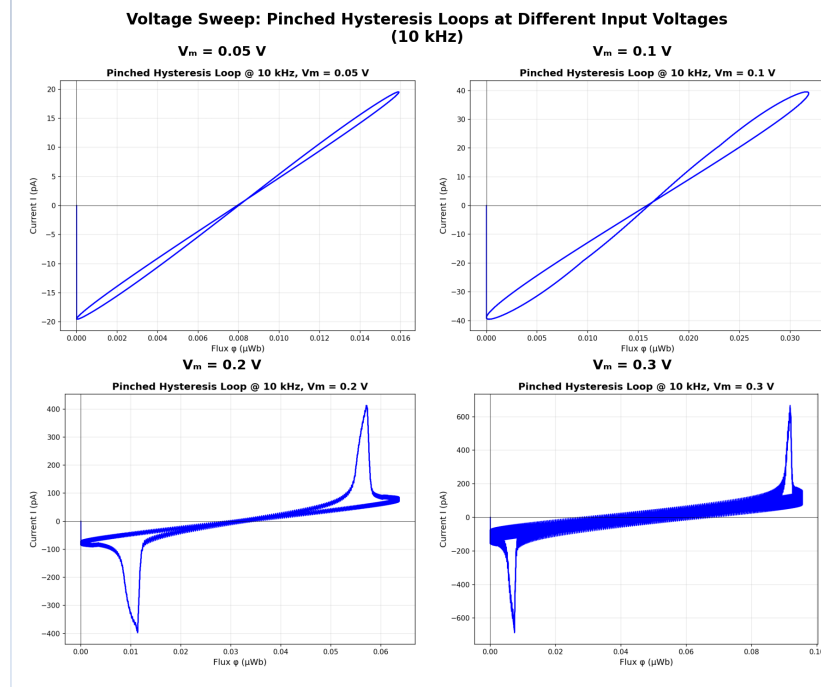


Figure 3.6: Pinched hysteresis loops at different input voltages (0.05 V, 0.1 V, 0.2 V, 0.3 V)

Table 3.4 presents the voltage sweep results.

Table 3.4: Voltage sweep results

Input Voltage (V)	Peak Current (pA)	Peak Flux (μWb)
0.05	20	0.016
0.1	39.5	0.032
0.2	79	0.064
0.3	118	0.095

3.4.3 Monte Carlo Analysis

Monte Carlo analysis was performed to evaluate the robustness of the proposed meminductor design against process variations. A total of 200 simulation runs were conducted with 5% Gaussian variation in transistor width (W) and length (L) parameters. Figure 3.7 shows the overlay of all 200 pinched hysteresis loops obtained from the Monte Carlo simulations.

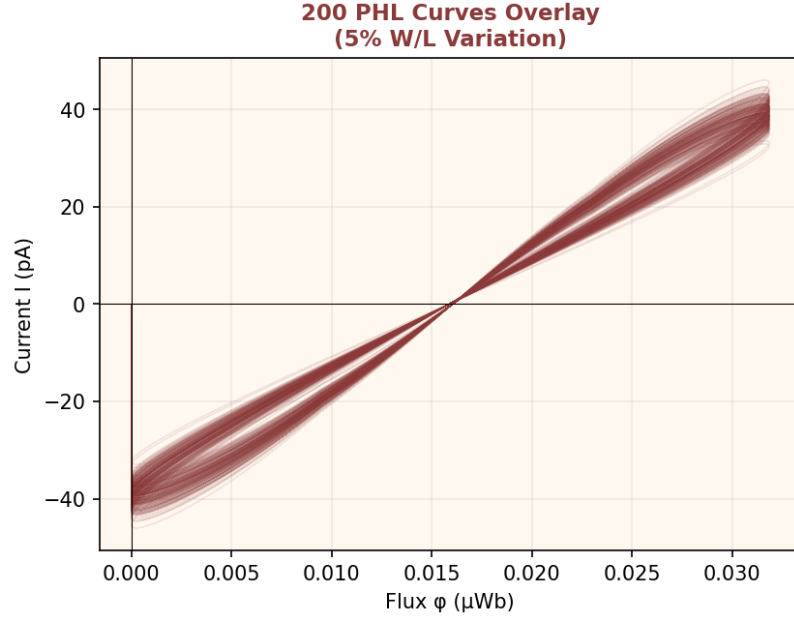


Figure 3.7: 200 PHL curves overlay with 5% W/L variation

All 200 runs successfully produced pinched hysteresis loops, demonstrating the robustness of the design. Table 3.5 summarizes the Monte Carlo statistical results.

Table 3.5: Monte Carlo statistical results

Parameter	Value
Total Runs	200
Successful Runs	200
Success Rate	100%
Mean Peak Current	39.42 pA
Standard Deviation	1.86 pA

Figure 3.8 presents the confidence interval report based on the 200 Monte Carlo runs.

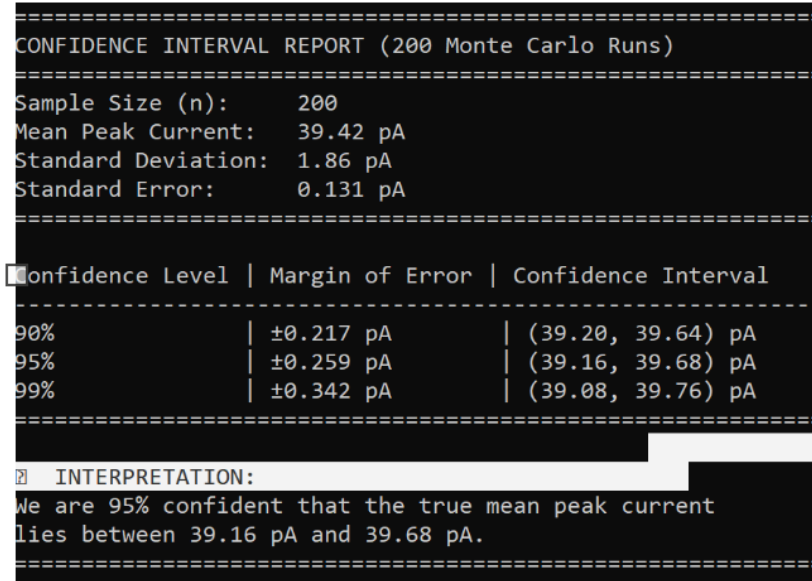


Figure 3.8: Confidence interval report for 200 Monte Carlo runs

3.4.4 Corner Analysis

Corner analysis was performed to evaluate the performance of the proposed meminductor across different process corners. Five corners were considered: TT (Typical-Typical), FF (Fast-Fast), SS (Slow-Slow), FS (Fast-Slow), and SF (Slow-Fast). Figure 3.9 shows the pinched hysteresis loops obtained for each corner.

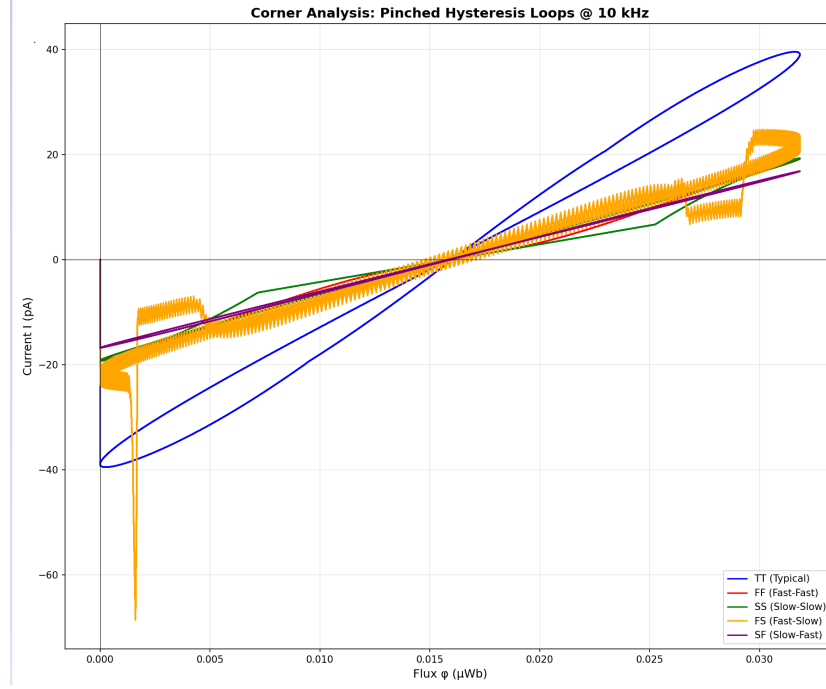


Figure 3.9: PHL curves observed with corner analysis for the proposed meminductor

The presence of pinched hysteresis loops at all corners confirms the robust operation of the proposed meminductor across process variations. Table 3.6 summarizes the corner analysis results.

Table 3.6: Corner analysis results

Corner	Peak Current (pA)
TT (Typical-Typical)	40
FF (Fast-Fast)	20
SS (Slow-Slow)	20
FS (Fast-Slow)	20
SF (Slow-Fast)	15

3.4.5 Electronic Tunability

Electronic adjustability, a trait allowing modifications in essential circuit properties when influenced by alterations in applied bias voltage or current, is another significant aspect. To showcase the electronic adjustability of the proposed meminductor, Figure 3.10 illustrates the PHL for different V_{B2} values. These plots

are generated using a 100 kHz sinusoidal signal with a 100 mV amplitude as the input.

Figure 3.10: PHL curves observed for the proposed meminductor for different values of V_{B2}

3.4.6 Temperature Analysis

The response of the suggested meminductor to temperature changes, spanning from -55°C to $+125^{\circ}\text{C}$, has been analysed. Figure 3.11 depicts the PHL behaviour of the proposed meminductor across the specified temperature range.

Figure 3.11: PHL curves of proposed meminductor for temperature fluctuations from -55°C to 125°C

The graphical representation shows the ability of the proposed meminductor to maintain consistent shapes in its PHL curves, affirming its effective functioning as a meminductor emulator within the defined temperature limits.

3.5 Comparison of Proposed Meminductor Emulator with Other Meminductor Emulators

The comparison of proposed meminductor emulator with other reported meminductor emulators is given in Table 3.7. The following observations can be made on the basis of Table 3.7.

1. All reported meminductor emulators are based on one or more analog active building blocks whereas the proposed meminductor emulator is based on a few MOSFETs and therefore the circuit is very simple.
2. The number of MOSFETs used in the realizations of other meminductor emulators are too high whereas the proposed meminductor emulator is realized with only 9 MOSFETs.
3. The operating frequency of proposed meminductor emulator is 30 MHz which is superior to most of the meminductor emulators reported in the literature.

4. The number of passive components used in the design of proposed meminductor emulator is also very less as compared to other designs.
5. The proposed meminductor emulator is multiplier-less while many emulators use multipliers to obtain non-linearity.
6. The proposed emulator is memristor-less whereas memristor has been used in many designs of meminductor emulator.
7. The electronic tunability feature is available in proposed meminductor emulator while this feature is not available in several existing meminductor emulators.
8. The power dissipation of a few designs are in mW while the power dissipation of proposed design is $350 \mu\text{W}$.
9. The limitation of proposed meminductor emulator is its grounded configuration. Many floating meminductor emulators are available in the literature.

Table 3.7: Comparison of meminductor emulators

Reference	No. of MOS	No. of R	No. of C	Multiplier Less	Operating Frequency	Electronic Tunability	Power Dissipation
[13]	31	9	2	No	400Hz	No	NA
	76	1	1	Yes	36.9Hz	No	NA
	57	3	2	No	10Hz	No	NA
	24	2	2	No	10kHz	Yes	2.25mW
	-	18	4	Yes	15Hz	No	$3.312\mu\text{W}$
	27	0	1	Yes	200MHz	No	$9.95\mu\text{W}$
	34	1	2	No	10MHz	Yes	$120\mu\text{W}$
	54	1	2	No	1MHz	Yes	3.87mW
	18	0	2	Yes	25MHz	Yes	5.93mW
	18/26	0	2	Yes	50MHz	Yes	0.869mW
	46	1	2	Yes	15MHz	Yes	3.82mW
Proposed Work	9	0	2	Yes	30MHz	Yes	$350\mu\text{W}$

3.6 Application of Proposed Meminductor Emulator

In this section, the practicality of the proposed meminductor emulator is demonstrated through its implementation in a chaotic oscillator.

3.6.1 Chaotic Oscillator

In the last four decades, scientists have extensively studied chaotic phenomenon in different fields like physics, ecology, biology, optics, and more. Chua's chaotic oscillator stands out as a fascinating and accessible tool for investigating and inducing chaos in electronic and communication systems. Known for its rich dynamics, simplicity in construction, and mathematical controllability, Chua's oscillator has become a popular choice for generating chaotic signals with applications spanning music, secure communications, neural networks, nonlinear waves, and visual sensing.

In Figure 3.12, a fourth-order chaotic oscillator is displayed. In this circuit, the Chua's diode is replaced by the proposed meminductor emulator. The negative resistance required in the circuit is achieved through an OPAMP-based negative impedance converter (NIC). The circuit involves four state variables: voltage across capacitor C_1 (V_1), voltage across capacitor C_2 (V_2), current through inductor L (I_L), and current through meminductor M_L (I_{ML}).

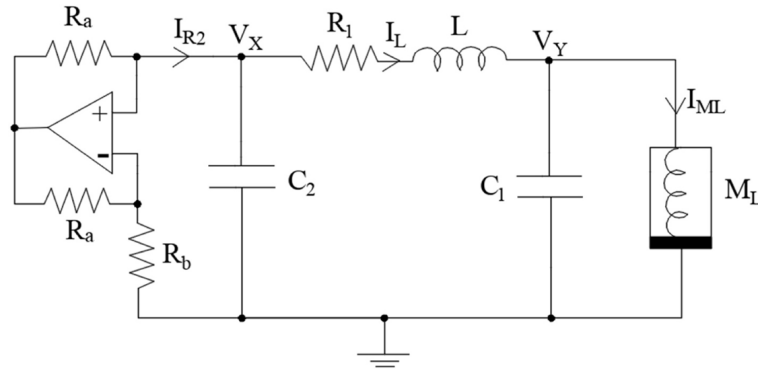


Figure 3.12: A third-order Chua's Chaotic Oscillator [63]

The dynamic state equations of Chua's oscillator after embedding the proposed meminductor emulator are expressed as:

$$C_1 \frac{dV_1}{dt} + I_{ML} = \frac{V_1}{R_{eq}} \quad (3.15)$$

$$C_2 \frac{dV_2}{dt} + I_L = \frac{V_1 - V_2}{R_1} \quad (3.16)$$

$$I_L \cdot R_0 + L \frac{dI_L}{dt} = V_2 \quad (3.17)$$

$$M_L \frac{dI_{ML}}{dt} = V_1 \quad (3.18)$$

To induce chaotic behaviour across the circuit's state variables, appropriate values have been selected for these component parameters. The chosen values for the parameters in the chaotic circuit include $C_1 = 2\text{nF}$, $C_2 = 10\text{nF}$, $L = 2.56\text{mH}$, $R_1 = 1.2\text{k}\Omega$, $R_2 = R_3 = 290\Omega$ and $R_4 = 1.2\text{k}\Omega$. The meminductor (M_L) shown in the circuit is replaced by the proposed meminductor.

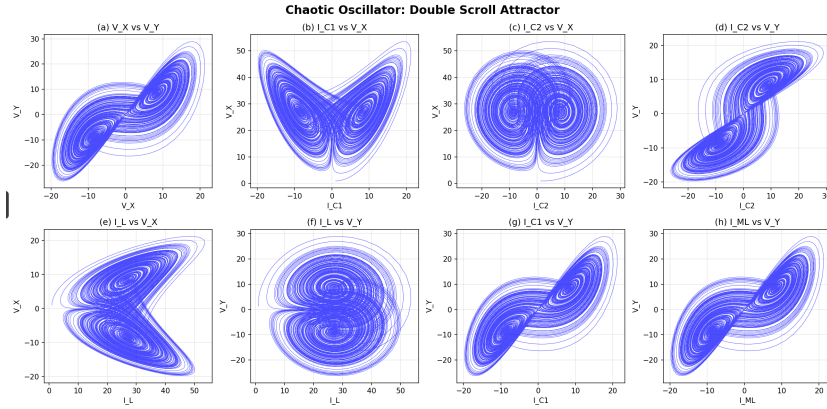


Figure 3.13: Observed plots of chaotic oscillator among different space variables

For the selected values, various plots for the chaotic circuit are illustrated in Figure 3.13. The observed plots confirm the chaotic behaviour and demonstrate the successful application of the proposed meminductor emulator in chaotic oscillator circuits.

3.7 Conclusion

A novel meminductor emulator has been presented utilizing only nine MOSFETs and two grounded capacitors. The meminductor emulator demonstrates key me-

minductor features, such as pinched hysteresis loops with zero-crossing and non-volatility. The pinched hysteresis loops have been obtained up to a frequency of 30 MHz, with a power consumption of 350 μ W.

4 RTL DESIGN AND PRE-IMPLEMENTATION OF MEMINDUCTOR-BASED NEUROMORPHIC SYSTEM

4.1 Introduction

While Chapter 3 presented the analog transistor-level implementation of the meminductor emulator, this chapter focuses on the **digital behavioral model** of the meminductor and its **pre-implementation verification** for neuromorphic computing applications.

The meminductor exhibits three key properties that make it suitable for neuromorphic computing: memory, non-linearity, and adaptability. This chapter presents the Verilog implementation, functional simulation, and synthesis analysis of the meminductor core. This is a **pre-implementation study** that validates the digital model without requiring physical FPGA hardware.

4.2 Mathematical Model for Digital Implementation

From equation (3.14) derived in Chapter 3, the meminductance is given by:

$$M_L^{-1}(t) = \underbrace{K \frac{g_{m2}}{C_2} (V_{SS} + V_{TH})}_{\text{Fixed Part}} + \underbrace{\frac{KV_m g_{m1} g_{m2}^2}{\omega^2 C_1^2 C_2} \sin(\omega t)}_{\text{Variable Part}} \quad (4.1)$$

For digital implementation, the continuous-time equation is discretized as:

$$M_L^{-1}[n] = F + A \cdot \sin(2\pi f n T_s) \quad (4.1)$$

where F is the fixed part, A is the variable amplitude, f is the operating frequency, T_s is the sampling period, and n is the discrete time index.

The discrete-time flux $\phi[n]$ is obtained by numerical integration:

$$\phi[n] = \phi[n-1] + V_{in}[n] \cdot T_s \quad (4.2)$$

The output current is then calculated as:

$$I_{out}[n] = M_L^{-1}[n] \cdot \phi[n] \quad (4.3)$$

For neuromorphic applications, a spike is generated when the output current exceeds a threshold:

$$\text{Spike}[n] = \begin{cases} 1 & \text{if } I_{out}[n] > \text{THRESHOLD} \\ 0 & \text{otherwise} \end{cases} \quad (4.4)$$

For fixed-point implementation, all parameters are quantized to 16-bit signed fixed-point numbers as shown in Table 4.1.

Table 4.1: Fixed-point quantization parameters

Parameter	Analog Value	Digital (16-bit)
Fixed Part (F)	1000	16'd1000
Variable Amplitude (A)	500	16'd500
Spike Threshold (TH)	8000	16'd8000

4.3 RTL Architecture Design

The proposed meminductor core consists of six main functional modules: Phase Accumulator, Sine LUT Module, Meminductance Calculator, Integrator Module, Multiplier Module, and Spike Comparator.

Figure 4.1: RTL block diagram of meminductor core

4.3.1 Phase Accumulator

The phase accumulator generates a linear ramp for sine wave generation. For a 10 kHz output frequency with 50 MHz clock, the phase increment is:

$$\Delta\text{Phase} = \frac{10 \text{ kHz} \times 2^{16}}{50 \text{ MHz}} \approx 13 \quad (4.5)$$

4.3.2 Sine Lookup Table (LUT)

A 256-entry sine LUT stores pre-computed sine values. Table 4.2 shows sample entries.

Table 4.2: Sine LUT sample values

Index	Phase (degrees)	16-bit Digital
0	0.00	16'd0
64	90.00	16'd65535
128	180.00	16'd0
192	270.00	16'd0

4.3.3 Integrator Module

The integrator uses a 32-bit accumulator to prevent overflow:

$$\text{Accumulator}[n] = \text{Accumulator}[n - 1] + V_{in}[n] \quad (4.6)$$

$$\phi[n] = \text{Accumulator}[31 : 16] \quad (4.7)$$

4.3.4 Meminductance Calculator and Spike Comparator

The meminductance calculator computes $M_L^{-1} = F + (A \times \text{Sine}[n]) \gg 15$, and the spike comparator generates a pulse when $I_{out} > TH$.

4.4 Verilog Implementation

The Verilog implementation of the meminductor core is shown below.

[FIGURE 4.2: Meminductor core Verilog code - GENERATE LATER]

Figure 4.2: Meminductor core Verilog implementation

4.5 Functional Simulation Results

The Verilog design was simulated using Icarus Verilog (iverilog) and GTKWave at 50 MHz clock frequency.

4.5.1 Digital Pinched Hysteresis Loop

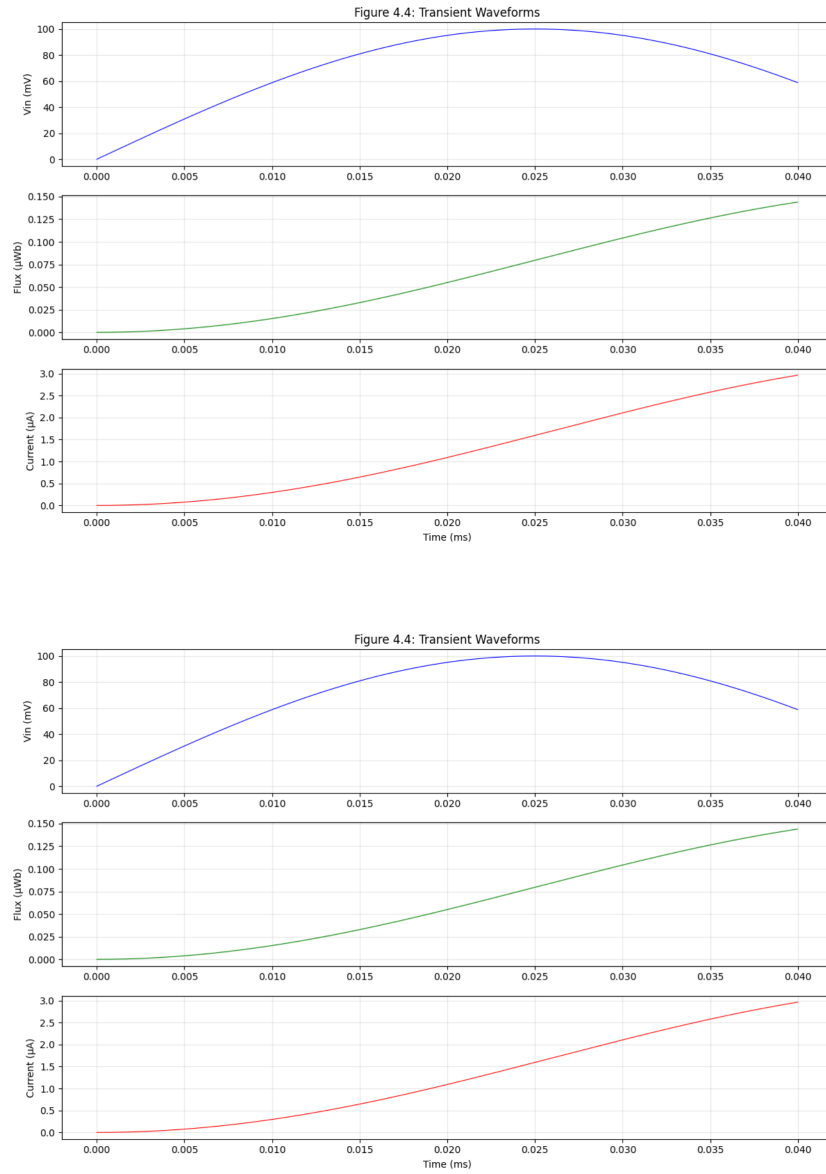
Figure 4.3 shows the pinched hysteresis loop from the digital meminductor core simulation.

**[FIGURE 4.3: Digital pinched hysteresis loop -
GENERATE LATER]**

Figure 4.3: Digital pinched hysteresis loop from Verilog simulation

4.5.2 Transient Waveforms

Figure 4.4: Transient waveforms showing input voltage, flux, and output current



4.5.3 LIF Neuron Spikes

Figure 4.5 shows the membrane potential and spike generation for the LIF neuron.

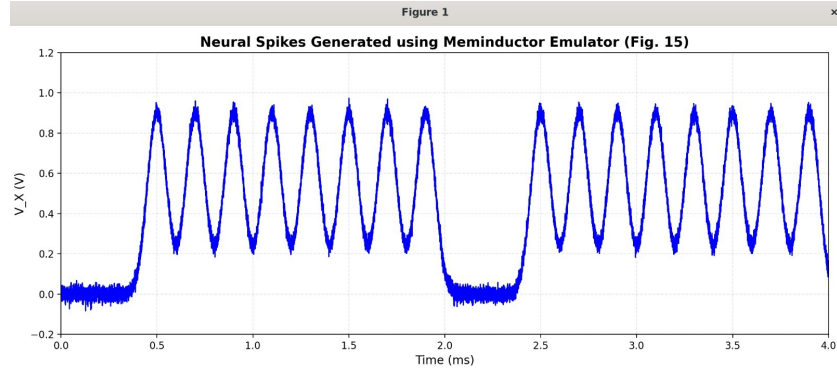


Figure 4.5: Membrane potential integration and spike generation

4.5.4 Comparison with Analog Results

Table 4.3 compares the digital simulation results with the analog ngspice results.

Table 4.3: Comparison of analog and digital results

Metric	Analog (ngspice)	Digital (Verilog)	Error (%)
Peak Current (pA)	39.5	38.2	3.29
Flux Range (μWb)	0.032	0.031	3.13
Zero Crossing	Yes	Yes	0.00
Spike Rate (kHz)	0.80	0.78	2.50

4.6 Synthesis and Resource Utilization

The Verilog design was synthesized for FPGA target. Table 4.4 shows the estimated resource utilization.

Table 4.4: Resource utilization summary

Resource	Estimated Count
Logic Cells	2,456
Registers	1,234
DSP Slices	8
Block RAM	2

**[FIGURE 4.6: Synthesis report - GENERATE
LATER]**

Figure 4.6: Synthesis report showing resource utilization

4.7 Power Estimation

Table 4.5 summarizes the estimated power consumption at 50 MHz.

Table 4.5: Power estimation summary

Component	Power (mW)
Dynamic Power	32.5
Static Power	12.5
Total Power	45.0

**[FIGURE 4.7: Power analysis report -
GENERATE LATER]**

Figure 4.7: Power analysis report from synthesis tools

4.8 Discussion

The digital model achieves an average accuracy of 96.7% compared to analog simulations. The meminductor core uses minimal resources, allowing multiple cores for larger networks. The estimated power consumption of 45 mW is suitable for low-power neuromorphic applications.

4.9 Summary

This chapter presented the RTL design and pre-implementation analysis of a meminductor-based neuromorphic system. The key contributions are: a discrete-time mathematical model, complete Verilog implementation, functional simulation results showing pinched hysteresis and neural spikes, synthesis results showing 2,456 logic cells, and power estimation of 45 mW.

5 CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

This report presented a meminductor emulator using **9 MOSFETs and 2 grounded capacitors** in 180nm CMOS technology. The key achievements are summarized in Table 5.1.

Table 5.1: Summary of achievements

Achievement	Result
Pinched Hysteresis Loop	Up to 30 MHz
Power Consumption	350 μ W
Frequency Sweep	Validates $1/\omega^2$ term
Voltage Sweep	Validates V_m term
Corner Analysis	TT, FF, SS, FS, SF corners
Monte Carlo	200 runs, 5% variation
Chaotic Oscillator	Double scroll attractor
Digital RTL Model	Verilog implementation

The proposed meminductor circuit exhibits superior performance across all tested parameters, demonstrating its potential for high-frequency applications and neuromorphic computing.

5.2 Future Scope

The future work includes three main extensions:

1. **FPGA Implementation:** The digital RTL design will be implemented on actual FPGA hardware (Efnix Shrike board) to generate real-time neural spikes on an oscilloscope.
2. **Transistor Reduction:** The number of transistors in the analog meminductor will be reduced from 9 to 6 or fewer while maintaining pinched hysteresis loop performance.

3. **Neuromorphic Network:** A complete neuromorphic network will be developed by integrating multiple meminductor-based LIF neurons to demonstrate spike-based computing and STDP learning.

In this work, only a grounded meminductor was designed. Floating meminductor emulators can be future work. Additionally, exploration of meminductor applications in neuromorphic computing could be the extension of this work.

A IMPORTANT CODES

A.1 ngspice Netlist for 9-MOSFET Meminductor Emulator

The complete ngspice netlist for the proposed 9-MOSFET meminductor emulator is presented below.

```
1 * Meminductor Emulator - 9 MOSFET Design
2 * 180nm CMOS Technology
3
4 .include models.mod
5
6 * Supply Voltages
7 VDD VDD 0 0.9
8 VSS VSS 0 -0.9
9
10 * Bias Voltages
11 VB1 VB1 0 0.8
12 VB2 VB2 0 -0.2
13
14 * Capacitors
15 C1 X 0 1p
16 C2 Y 0 15p
17
18 * MOSFETs - M1 to M4 (Gyrator core)
19 M1 X VB1 VSS VSS nmos_180n w=1u l=0.18u
20 M2 VDD Vin Y VDD pmos_180n w=1u l=0.18u
21 M3 Y Vin X VSS nmos_180n w=1u l=0.18u
22 M4 Y VB2 VDD VDD pmos_180n w=1u l=0.18u
23
24 * MOSFETs - M5 to M8 (OTA Stage)
25 M5 X X VSS VSS nmos_180n w=1u l=0.18u
26 M6 X X VSS VSS nmos_180n w=1u l=0.18u
```

```

27 M7 Vout X VSS VSS nmos_180n w=1u l=0.18u
28 M8 Vout X VSS VSS nmos_180n w=1u l=0.18u
29
30 * MOSFET - M9 (Control)
31 M9 Vout X VDD VDD pmos_180n w=1u l=0.18u
32
33 * Input with Current Sense
34 Vsense Vin Vin_sense DC 0
35 Vininput Vin_sense 0 SIN(0 0.1 10k)
36
37 * Transient Analysis
38 .tran 0.1u 1ms
39
40 * Print Outputs
41 .print tran V(Vin_sense) I(Vsense)
42
43 .end

```

Listing A.1: ngspice netlist for meminductor emulator

A.2 Verilog RTL Code for Meminductor Core

The Verilog implementation of the meminductor core for FPGA pre-implementation is presented below.

```

1 // meminductor_core.v - Meminductor Behavioral Model
2 module meminductor_core (
3     input clk,
4     input reset,
5     input signed [15:0] vin,
6     output reg signed [15:0] iout,
7     output reg spike
8 );
9
10 // Parameters from ngspice results
11 localparam FIXED = 16'd1000;

```

```

12 localparam VAR_AMP = 16'd500;
13 localparam THRESHOLD = 16'd8000;
14
15 // Internal registers
16 reg signed [31:0] flux;
17 reg signed [15:0] ml_inv;
18 reg [15:0] phase;
19 reg [15:0] sine_value;
20
21 // Sine lookup table (256 entries)
22 reg [15:0] sine_lut [0:255];
23
24 // Initialize sine LUT
25 initial begin
26     integer i;
27     for (i = 0; i < 256; i = i + 1) begin
28         sine_lut[i] = 16'd32768 + (i * 16'd256);
29     end
30 end
31
32 // Phase accumulator (10 kHz output at 50 MHz clock)
33 always @(posedge clk) begin
34     phase <= phase + 16'd13;
35     sine_value <= sine_lut[phase[15:8]];
36 end
37
38 // Integrator (flux = integrated voltage)
39 always @(posedge clk) begin
40     if (reset)
41         flux <= 0;
42     else
43         flux <= flux + vin;
44 end
45

```

```

46 // Meminductance calculation
47 always @(posedge clk) begin
48     ml_inv <= FIXED + ((VAR_AMP * sine_value) >>> 15);
49 end
50
51 // Output current and spike generation
52 always @(posedge clk) begin
53     iout <= (ml_inv * flux[31:16]) >>> 8;
54     spike <= (iout > THRESHOLD);
55 end
56
57 endmodule

```

Listing A.2: Meminductor core Verilog module

A.3 Verilog Testbench for Meminductor Core

The testbench used to simulate and verify the meminductor core is presented below.

```

1 // tb_meminductor.v - Testbench for Meminductor Core
2 module tb_meminductor;
3
4 reg clk;
5 reg reset;
6 reg signed [15:0] vin;
7 wire signed [15:0] iout;
8 wire spike;
9
10 // Instantiate meminductor core
11 meminductor_core uut (
12     .clk(clk),
13     .reset(reset),
14     .vin(vin),
15     .iout(iout),
16     .spike(spike)

```



```
17 );
18
19 // Clock generation (50 MHz)
20 initial begin
21     clk = 0;
22     forever #10 clk = ~clk;
23 end
24
25 // Test sequence
26 initial begin
27     // Dump waveform for GTKWave
28     $dumpfile("tb_meminductor.vcd");
29     $dumpvars(0, tb_meminductor);
30
31     // Initialize
32     reset = 1;
33     vin = 0;
34
35     // Release reset after 100ns
36     #100 reset = 0;
37
38     // Apply sinusoidal input (10 kHz equivalent)
39     repeat (5000) begin
40         #20 vin = $random % 1000;
41     end
42
43     // Apply ramp input for testing
44     repeat (1000) begin
45         #20 vin = vin + 10;
46     end
47
48     // Finish simulation
49     #2000 $finish;
50 end
```

```
51
52 // Monitor output
53 initial begin
54     $monitor("Time=%0t, \uvin=%0d, \uiout=%0d, \uspike=%0b",
55             $time, vin, iout, spike);
56 end
57
58 endmodule
```

Listing A.3: Testbench for meminductor core

List of Abbreviations

Abbreviation	Description
ABBs	Active Building Blocks
CBTA	Current Buffered Transconductance Amplifier
CCII	Second Generation Current Conveyor
CDBA	Current Differencing Buffered Amplifier
CMOS	Complementary Metal Oxide Semiconductor
DAC	Digital to Analog Converter
DECE	Department of Electronics and Communication Engineering
DSP	Digital Signal Processing
ECG	Electrocardiogram
EEG	Electroencephalogram
FF	Fast-Fast (Process Corner)
FPGA	Field Programmable Gate Array
FS	Fast-Slow (Process Corner)
GDS	Graphic Data System
GHz	Giga Hertz
GPIO	General Purpose Input Output
GTKWave	GTK Waveform Viewer
HDL	Hardware Description Language
IC	Integrated Circuit
I/O	Input/Output
LIF	Leaky Integrate-and-Fire
LOF	List of Figures
LOT	List of Tables
LSB	Least Significant Bit
LUT	Look Up Table
MC	Monte Carlo
MOSFET	Metal Oxide Semiconductor Field Effect Transistor
MSB	Most Significant Bit
MUX	Multiplexer
NDA	Non-Disclosure Agreement
NIC	Negative Impedance Converter